

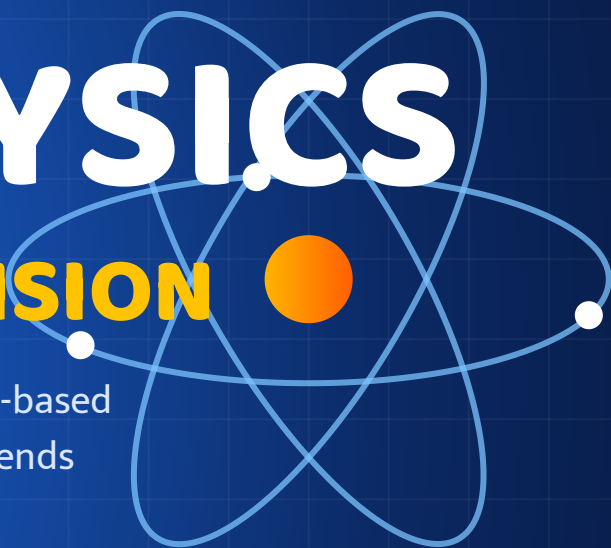
CLASS 11 · NCERT ALIGNED · NEET + JEE MAIN

$$v^2 = u^2 + 2as$$

NEET PHYSICS

30-DAY RAPID REVISION

Concepts · Formulas · Diagrams · Traps · Pattern-based practice from NEET & JEE Main previous-year trends



$$E = \frac{1}{2}mv^2$$

$$\tau = I\alpha$$

14 NCERT Chapters

300+ MCQs with Solutions

Full Mock Test

100+ Diagrams & Formula Boxes

$T =$



VS SIR

Vijay Kumar Sharma
20+ years · NEET/JEE Physics · Kota

NEET Physics: 30-Day Rapid Revision

Class 11 Physics · NCERT-aligned · For NEET & JEE Main

by VS Sir (Vijay Kumar Sharma)

AI-Enabled Educator · 20+ years teaching NEET/JEE Physics, Kota

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About the questions: All practice questions in this book are original, pattern-based questions written by the author and modelled on the trends of NEET and JEE Main previous-year papers. They are not reproductions of official papers. Syllabus follows the rationalised NCERT Class 11 textbook and the NMC NEET-UG syllabus.

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ABOUT THE AUTHOR

VS Sir

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- 20+ years of teaching Physics to NEET and JEE aspirants in Kota
- AI-Enabled Educator – trains teachers to use AI tools in the classroom
- Author of "Practical Guide to AI for Educators", "Electrostatics" and "Automation Mastery"
- Known for turning long chapters into one-page concept maps and formula boxes

A message from VS Sir

Dear student,

In the last month before NEET, you do not need another 800-page book. You need a **clear, short, reliable revision** of everything NCERT says, the formulas that actually get used, and the traps where toppers lose marks.

This book is built exactly for that. One chapter per day or two, a formula box you can revise in five minutes, diagrams that make the idea obvious, and practice questions in the pattern NEET and JEE Main have followed year after year – each with a quick solution.

Follow the 30-day plan honestly, tick each box, and keep an error log. Consistency beats intensity. You can do this!

— VS Sir

How to use this book

Every chapter follows the same rhythm so your brain knows exactly where to look.

1. **Read** the chapter-at-a-glance map (2 min).
2. **Revise** concepts, diagrams and formula boxes (40–60 min).
3. **Solve** the 20 practice questions with a timer – 1 min each.
4. **Check** with the answer key; read the solution of every wrong/guessed question.
5. **Log** each mistake in your error log (page at the end). Re-solve on Day 29.

Colour code of the boxes

 Key Concept Core idea explained simply	 Formula Box Formulas to revise in 5 minutes
 Common Trap Where students lose marks	 VS Sir's Exam Tip Exam strategy & shortcuts
 NCERT Line to Remember Exact NCERT facts asked in NEET	 Memory Trick Tricks to remember
 JEE Main Extension Extra depth for JEE Main	

Question tags

NCERT based direct from textbook lines **NEET pattern** typical NEET level **JEE Main pattern** one step deeper
Assertion–Reason

 Honest note on PYQs The questions are original and written in the style of NEET and JEE Main previous-year questions (PYQ trends): same concepts, same difficulty, same traps. Solve your official PYQ papers too – this book prepares you to crack them.

Your 30-Day Revision Plan

Daily target: 2–3 hours for Physics. Tick the box when a day is complete. Weekly tests are built into Days 8 and 17.

Day 1 Ch 1 Units, SF, errors, dimensions	Day 2 Ch 2 Motion in a straight line	Day 3 Ch 3 Vectors & projectile motion	Day 4 Ch 3 Relative velocity, circular motion	Day 5 Ch 4 Newton's laws, FBDs
Day 6 Ch 4 Friction, circular dynamics	Day 7 Ch 5 Work, energy, power	Day 8 Ch 5 Collisions + Test (Ch 1–5)	Day 9 Ch 6 Centre of mass, torque	Day 10 Ch 6 Moment of inertia, rolling
Day 11 Ch 7 Kepler's laws, g variation	Day 12 Ch 7 Satellites, escape speed	Day 13 Ch 8 Elasticity of solids	Day 14 Ch 9 Pressure, Pascal, buoyancy	Day 15 Ch 9 Bernoulli, viscosity, surface tension
Day 16 Ch 10 Expansion, calorimetry	Day 17 Ch 10 Heat transfer + Test (Ch 6–10)	Day 18 Ch 11 Laws of thermodynamics	Day 19 Ch 11 Processes, engines, Carnot	Day 20 Ch 12 Kinetic theory of gases
Day 21 Ch 13 SHM basics & energy	Day 22 Ch 13 Spring & pendulum	Day 23 Ch 14 Wave motion, wave speed	Day 24 Ch 14 Standing waves, pipes, beats	Day 25 Final Formula sheet: Mechanics
Day 26 Final Formula sheet: Rest of Class 11	Day 27 Final FULL MOCK TEST (45 Q, 60 min)	Day 28 Final Mock analysis + error log	Day 29 Final Re-attempt all wrong questions	Day 30 Final Traps list + exam-day plan

Golden rules

- Never skip the practice set – revision without questions is only reading.
- Sleep 7 hours. Memory is stored while you sleep.

If you fall behind

Do the formula box + 10 questions of the missed day, then continue with today. Use Days 28–29 as buffer.

Contents

1	Units and Measurement SI units · Significant figures · Errors · Least count	Day 1	7
2	Motion in a Straight Line Displacement vs distance · Speed & velocity · Acceleration · x - t , v - t , a - t graphs	Day 2	14
3	Motion in a Plane Scalars & vectors · Vector addition · Dot & cross product · Projectile motion	Day 3–4	22
4	Laws of Motion Newton's laws · Momentum & impulse · Free-body diagrams · Apparent weight	Day 5–6	31
5	Work, Energy and Power Work · Work–energy theorem · Potential energy · Conservation of energy	Day 7–8	41
6	System of Particles and Rotational Motion Centre of mass · Torque & angular momentum · Moment of inertia · Axis theorems	Day 9–10	50
7	Gravitation Kepler's laws · Law of gravitation · Variation of g · Potential & PE	Day 11–12	61
8	Mechanical Properties of Solids Stress & strain · Hooke's law · Stress–strain curve · Y , B , G moduli	Day 13	70
9	Mechanical Properties of Fluids Pressure & Pascal · Continuity · Bernoulli's principle · Viscosity & Stokes	Day 14–15	78
10	Thermal Properties of Matter Temperature scales · Thermal expansion · Calorimetry · Latent heat	Day 16–17	87
11	Thermodynamics Zeroth & first law · C_p , C_v , γ · Isothermal & adiabatic · Work = area on PV	Day 18–19	96
12	Kinetic Theory Ideal gas equation · Pressure of a gas · KE & temperature · rms speed	Day 20	105
13	Oscillations Periodic motion · SHM equation · Velocity & acceleration · Energy in SHM	Day 21–22	113
14	Waves Transverse & longitudinal · Wave equation · Speed of waves · Superposition & reflection	Day 23–24	123
	Full Mock Test 45 questions · 60 minutes · NEET marking	Day 27–29	133
	Last-Week Toolkit: Formula Sheets, Traps & Exam Strategy 14 formula sheets · 60 traps · 10 key graphs · exam-day plan · error log	Day 25–30	141

CHAPTER

1

Units and Measurement

17

Day 1

NEET weightage ★★☆☆

NCERT Class 11

SI units

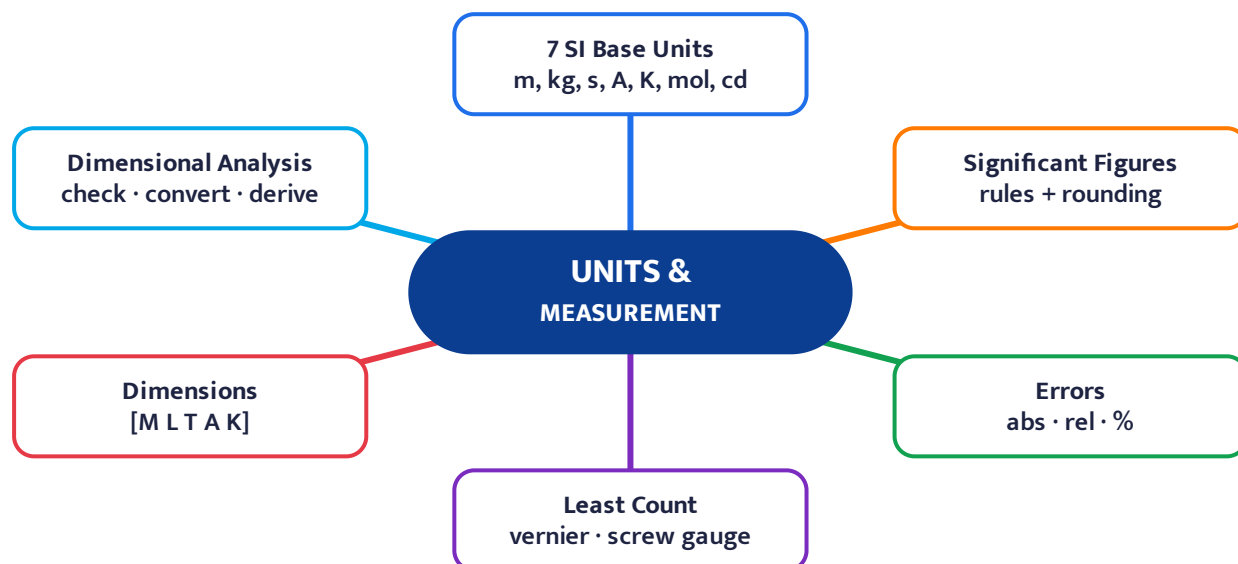
Significant figures

Errors

Least count

Dimensions

Dimensional analysis



Chapter at a glance

1.1 The SI System – 7 Base Quantities

Every physical quantity is expressed as **(numerical value) × (unit)**. The International System (SI) builds all units from **seven base units**; every other unit is a **derived unit**.

Base quantity	SI unit	Symbol	Dimension
Length	metre	m	[L]
Mass	kilogram	kg	[M]
Time	second	s	[T]
Electric current	ampere	A	[A]
Thermodynamic temperature	kelvin	K	[K]
Amount of substance	mole	mol	[mol]
Luminous intensity	candela	cd	[cd]

 NCERT Line to Remember

Plane angle (radian, rad) and solid angle (steradian, sr) are **dimensionless**. $d\theta = \frac{ds}{r}$ and $d\Omega = \frac{dA}{r^2}$. A full sphere subtends 4π sr.

 Memory Trick

Prefixes you must know: **p** 10^{-12} , **n** 10^{-9} , **μ** 10^{-6} , **m** 10^{-3} , **c** 10^{-2} , **k** 10^3 , **M** 10^6 , **G** 10^9 , **T** 10^{12} .

 Handy practical units

- 1 light year = 9.46×10^{15} m
- 1 AU = 1.496×10^{11} m
- 1 parsec = 3.08×10^{16} m $\approx$ 3.26 ly
- 1 Å = 10^{-10} m, 1 fermi = 10^{-15} m
- 1 u = 1.66×10^{-27} kg

1.2 Significant Figures

Significant figures (SF) are the **reliable digits plus the first uncertain digit** of a measurement. They tell you how precise a reading is.

Rule	Example	SF
All non-zero digits are significant	2.345	4
Zeros between non-zero digits count	20.05	4
Leading zeros (before first non-zero) do NOT count	0.00450	3
Trailing zeros after decimal point count	4.500	4
Trailing zeros without decimal: ambiguous – use scientific notation	4.50×10^3	3
Powers of 10 are never counted	6.0×10^{23}	2

 Arithmetic with SF

- **× and ÷** : answer has as many SF as the quantity with the **least SF**.
- **+** and **–** : answer has as many **decimal places** as the quantity with the **fewest decimal places**.

 Rounding off

- Digit dropped $> 5 \rightarrow$ raise; $< 5 \rightarrow$ keep.
- Exactly 5: make preceding digit **even**. 2.745 $\rightarrow$ 2.74, 2.735 $\rightarrow$ 2.74.

 Solved Example

Find 4.237×2.51 and $12.11 + 18.0 + 1.012$ to correct significant figures.

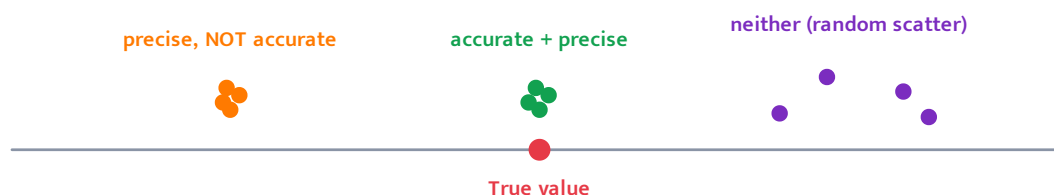
Solution: Product = 10.63487 $\rightarrow$ least SF is 3 (2.51) $\rightarrow$ **10.6**. Sum = 31.122 $\rightarrow$ fewest decimal places is 1 (18.0) $\rightarrow$ **31.1**.

 Common Trap

- Changing units does **not** change SF: 2.30 m = 230 cm = 2.30×10^2 cm (3 SF each).
- 0.0500 has **3** SF, not 1 – the trailing zeros after the decimal count.
- Exact numbers (like 2 in $2\pi r$) have infinite SF – ignore them.

1.3 Errors in Measurement

Accuracy = closeness to the true value. **Precision** = closeness of repeated readings to each other (resolution). Errors are **systematic** (instrument, method, personal – same sign every time) or **random** (irregular, reduced by taking many readings).



Accuracy vs precision – the favourite NCERT picture question

Error formulas

$\bar{a} = \frac{a_1 + a_2 + \dots + a_n}{n}$	Mean (best) value
$\Delta a_i = a_i - \bar{a}$	Absolute error of each reading
$\bar{\Delta a} = \frac{\sum \Delta a_i }{n}$	Mean absolute error
$\delta a = \frac{\bar{\Delta a}}{\bar{a}}$	Relative (fractional) error
$\frac{\bar{\Delta a}}{\bar{a}} \times 100\%$	Percentage error

Combination of errors (always ADD the errors)

Propagation of errors

$Z = A \pm B \Rightarrow \Delta Z = \Delta A + \Delta B$	Sum or difference: absolute errors add
$Z = AB \text{ or } \frac{A}{B} \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$	Product or quotient: relative errors add
$Z = \frac{A^p B^q}{C^r} \Rightarrow \frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$	Powers multiply the relative error

Solved Example

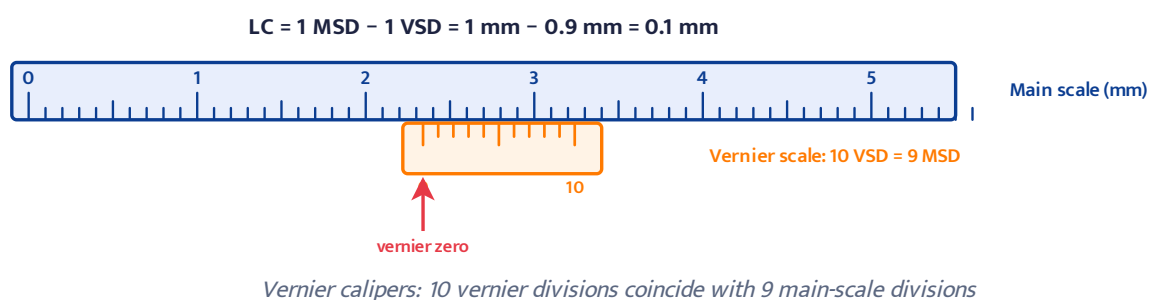
The period of a pendulum is $T = 2\pi\sqrt{l/g}$. If l is measured with 2% error and T with 1% error, find the % error in g .

Solution: $g = 4\pi^2 l / T^2 \Rightarrow \frac{\Delta g}{g} = \frac{\Delta l}{l} + 2 \frac{\Delta T}{T} = 2\% + 2(1\%) = 4\%$.

VS Sir's Exam Tip

In NEET, the quantity with the **highest power** contributes most to the error – measure it most carefully. Signs in the formula never reduce error: $\frac{\Delta Z}{Z}$ for $Z = A / B$ is still a **sum**.

1.4 Least Count – Vernier Calipers & Screw Gauge



Least-count formulas

$$LC_{\text{vernier}} = 1 \text{ MSD} - 1 \text{ VSD}$$

Least count of vernier calipers

$$\text{Reading} = \text{MSR} + (\text{VSR} \times \text{LC}) - \text{zero error}$$

Vernier reading

$$LC_{\text{screw}} = \frac{\text{pitch}}{\text{no. of circular divisions}}$$

Screw gauge least count (e.g. $0.5 \text{ mm}/50 = 0.01 \text{ mm}$)

Common Trap

Zero error is **subtracted with its sign**: positive zero error (+0.02 cm) → subtract 0.02; negative zero error (−0.03 cm) → add 0.03.

1.5 Dimensions & Dimensional Analysis

The **dimensions** of a quantity are the powers to which the base quantities are raised to represent it. Example: force $F = ma \Rightarrow [F] = [M^1 L^1 T^{-2}]$.

Quantity	Dimension	Quantity	Dimension
Velocity	$[LT^{-1}]$	Pressure, stress, Y	$[ML^{-1}T^{-2}]$
Force	$[MLT^{-2}]$	Surface tension, spring k	$[MT^{-2}]$
Work, energy, torque	$[ML^2T^{-2}]$	Viscosity η	$[ML^{-1}T^{-1}]$
Power	$[ML^2T^{-3}]$	Gravitational constant G	$[M^{-1}L^3T^{-2}]$
Momentum, impulse	$[MLT^{-1}]$	Planck constant h, angular momentum	$[ML^2T^{-1}]$
Moment of inertia	$[ML^2]$	Specific heat c	$[L^2T^{-2}K^{-1}]$
Strain, angle, refractive index	dimensionless	Gas constant R	$[ML^2T^{-2}K^{-1}mol^{-1}]$

 3 uses

- **Check** correctness of an equation (principle of homogeneity).
- **Convert** units: $n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$
- **Derive** a relation between quantities (up to a constant).

 Limitations

- Cannot find dimensionless constants (2π , $\frac{1}{2}$).
- Fails for sin, log, exp relations and sums of terms.
- A dimensionally correct equation may still be wrong.
- Fails if more than 3 base quantities are involved (M, L, T).

 Solved Example

Using dimensions, show that the time period of a simple pendulum is $T = k\sqrt{l/g}$.

Solution: Let $T = k m^a l^b g^c$. Then $[T] = [M]^a [L]^b [LT^{-2}]^c$. Equate powers: M: $a = 0$; L: $b + c = 0$; T: $-2c = 1 \Rightarrow c = -\frac{1}{2}, b = \frac{1}{2}$. So $T = k\sqrt{l/g}$ (experiment gives $k = 2\pi$).

 Solved Example

Convert 1 newton into dyne using dimensions.

Solution: $[F] = [MLT^{-2}]$. $n_2 = 1 \times \left(\frac{1\text{ kg}}{1\text{ g}} \right)^1 \left(\frac{1\text{ m}}{1\text{ cm}} \right)^1 \left(\frac{1\text{ s}}{1\text{ s}} \right)^{-2} = 10^3 \times 10^2 = 10^5$. So **1 N = 10^5 dyne**.

 Memory Trick

Quick pairs with the SAME dimensions: work–torque–energy; pressure–stress–Young's modulus–energy density; impulse–momentum; Planck constant–angular momentum; frequency–angular velocity–velocity gradient.

 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

1

NCERT based The number of significant figures in 0.06900 is

(a) 3

(b) 4

(c) 5

(d) 2

2

NEET pattern Which pair has the same dimensions?

(a) Momentum and energy

(b) Torque and energy

(c) Work and power

(d) Force and pressure

3

NEET pattern The dimensions of Planck's constant are the same as those of

(a) energy

(b) linear momentum

(c) angular momentum

(d) power

4

NEET pattern A physical quantity $P = \frac{a^3 b^2}{\sqrt{c} d}$. Percentage errors in a, b, c, d are 1%, 3%, 4% and 2%. The % error in P is

(a) 7%

(b) 10%

(c) 13%

(d) 14%

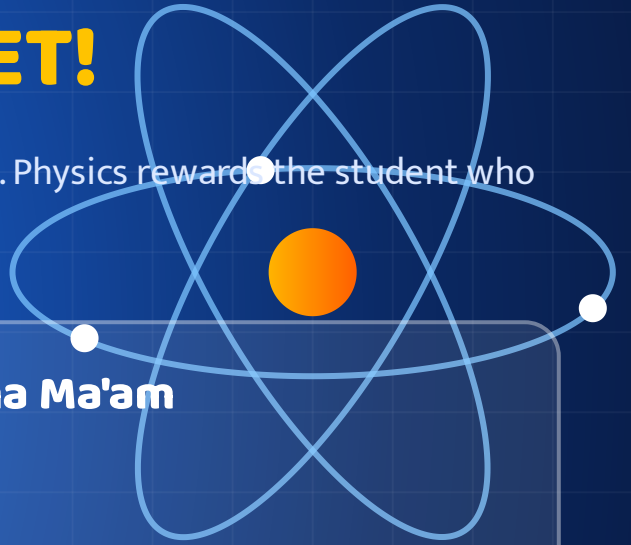
- 5 **NCERT based** The dimensional formula of the coefficient of viscosity is
 (a) $[MLT^{-1}]$ (b) $[M^0L^0T^0]$ (c) $[ML^{-1}T^{-1}]$ (d) $[ML^{-2}T^{-2}]$
- 6 **JEE Main pattern** In a vernier calipers, 20 vernier divisions coincide with 19 main scale divisions of 1 mm each. Its least count is
 (a) 0.02 mm (b) 0.05 mm (c) 0.1 mm (d) 0.5 mm
- 7 **NCERT based** The sum of 2.745, 1.32 and 0.004 with correct significant figures is
 (a) 4.069 (b) 4.0 (c) 4.1 (d) 4.07
- 8 **NEET pattern** If force F , velocity v and time T are taken as fundamental quantities, the dimensions of mass are
 (a) $[Fv^{-1}T]$ (b) $[FvT^{-1}]$ (c) $[F^{-1}vT]$ (d) $[FvT]$
- 9 **JEE Main pattern** A screw gauge has pitch 0.5 mm and 50 circular divisions. Main scale reading 2.5 mm, 20th circular division coincides, zero error -0.01 mm. The reading is
 (a) 2.70 mm (b) 2.71 mm (c) 2.69 mm (d) 2.20 mm
- 10 **NCERT based** Which of the following is dimensionless?
 (a) Strain (b) Surface tension
 (c) Stress (d) Angular velocity
- 11 **NEET pattern** The radius of a sphere is measured as (2.0 ± 0.1) cm. The percentage error in its volume is
 (a) 5% (b) 10% (c) 20% (d) 15%
- 12 **NEET pattern** In $y = A\sin(\omega t - kx)$, the dimensions of ω / k are
 (a) $[LT^{-1}]$ (b) $[L^{-1}]$ (c) $[L^{-1}T]$ (d) $[T^{-1}]$
- 13 **NCERT based** The dimensional formula of the universal gravitational constant G is
 (a) $[M^{-1}L^2T^{-2}]$ (b) $[ML^3T^{-2}]$ (c) $[ML^2T^{-2}]$ (d) $[M^{-1}L^3T^{-2}]$
- 14 **JEE Main pattern** If the unit of length is doubled and unit of mass is halved (time unchanged), the numerical value of a given force becomes
 (a) same (b) half (c) double (d) four times
- 15 **Assertion-Reason** Assertion: A dimensionally correct equation must be physically correct. Reason: Dimensional analysis cannot detect missing dimensionless constants.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 16 **NCERT based** Two resistors $R_1 = (100 \pm 3) \Omega$ and $R_2 = (200 \pm 4) \Omega$ are in series. The equivalent resistance is
 (a) $(300 \pm 5) \Omega$ (b) $(300 \pm 1) \Omega$ (c) $(300 \pm 7) \Omega$ (d) $(300 \pm 12) \Omega$

$$v^2 = u^2 + 2as$$

All the best for NEET!

$$F = ma$$

Revise daily, test weekly, analyse every mistake. Physics rewards the student who practises smartly.



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$$E = \frac{1}{2}mv^2$$

$$\tau = I\alpha$$

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$$PV = nRT$$

$$T = 2\pi\sqrt{l/g}$$

